Foundation Models for PDEs in fluid dynamics

M2 Master's Thesis

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Introduction: Numerical Methods for solving PDEs

Partial Differential Equations (PDEs): Fundamental mathematical models that describe a wide variety of physical phenomena

Traditional Numerical Methods:

- Finite difference, finite element, spectral methods
- Approximate the PDE solution on discretized space and time domain
- Simplify the initial problem to solving a system of algebraic equations
- **Limitations:** Iterative solving, high (prohibitive) computational costs e.g. for many-query problems

Data-driven Machine Learning (ML) Methods:

- **Aim (Operator Learning):** Learn the underlying PDE solution operator $\mathcal{S}$
- Significant advantages in terms of computational efficiency, robustness and accuracy
- Allows to predict the entire solution trajectory in a single forward pass

Operator Learning

Operator learning task: Given the generic time-dependent PDE

$$\partial_t u(x,t) + \mathcal{L}\left(u, \nabla_x u, \nabla_x^2 u, \ldots\right) = 0, \quad \forall x \in D \subset \mathbb{R}^d, t \in (0,T),$$
$$\mathcal{B}(u) = 0, \quad \forall (x,t) \in \partial D \times (0,T), \quad u(0,x) = a(x), \quad x \in D$$

where:

- $D = [0,1]^2$ is the spatial domain
- $u \in C([0,T]; \mathcal{X})$ is the solution with $\mathcal{X} \subset C(D; \mathbb{R}^n)$
- $\mathcal{L}, \mathcal{B}$ are differential and boundary operators

PDE solution operators are treated as **function-to-function mappings** in continuous function spaces $\mathcal{X} \subset C(D; \mathbb{R}^n)$, mapping function space inputs (initial and boundary conditions, coefficients, sources) to the PDE solution.

Goal: Learn an approximation $\mathcal{S}^* \approx \mathcal{S}$ of the solution operator $\mathcal{S} : [0,T] \times \mathcal{X} \mapsto \mathcal{X}$ allowing to generate the entire solution trajectory $\{\mathcal{S}^*(t,a)\}$ for all $t \in [0,T]$.

POSEIDON : Foundation Model for PDEs

Motivation:

How can the number of training samples for PDE learning be significantly reduced?

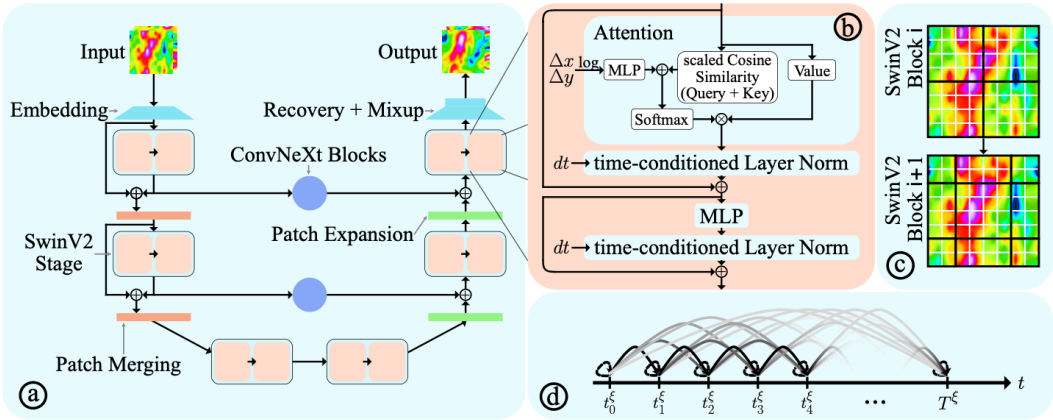
all2all Training Strategy: This considers all possible pairs of data points $(u(t_k), u(t_{\bar{k}}))$ within the trajectory where $k \neq \bar{k}$.
This increases the number of training pairs from $O(K)$ to $O(K^2)$ per trajectory while maintaining physical consistency of the underlying PDE dynamics.

Foundation Model Approach: Pretraining on a diverse set of PDEs and underlying data-distributions; fine-tuning on a small number of **task-specific** training examples.

Open Question:

Why can pretraining enable generalization to **unseen and unrelated** PDEs via finetuning?

POSEIDON Model Architecture



(a-b) Scalable Operator Transformer (scOT) architecture and (d) **all2all** training strategy. Figure taken from: Herde et al. [1]

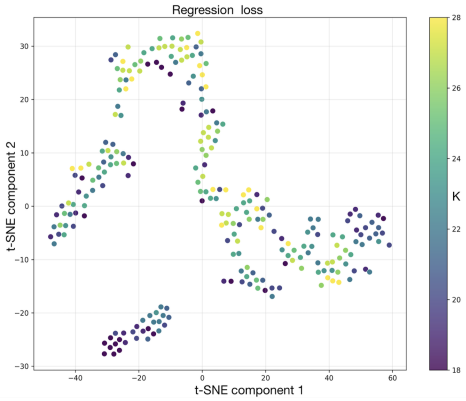
Motivation and Research Objectives

Current Approach:

- POSEIDON uses an L_1 loss
- This ignores the structure of the target space leading to **fragmented representations**

Motivation:

- **Representation Learning:** Capture governing dynamics and data relationships; improved downstream performance



Latent space representation of POSEIDON

Research Goal:

Implement a learning approach explicitly emphasizing the learning of **regression-aware representations** to capture the **continuous nature of the target space**.

Rank-N-Contrast (RNC): Contrastive Learning Framework

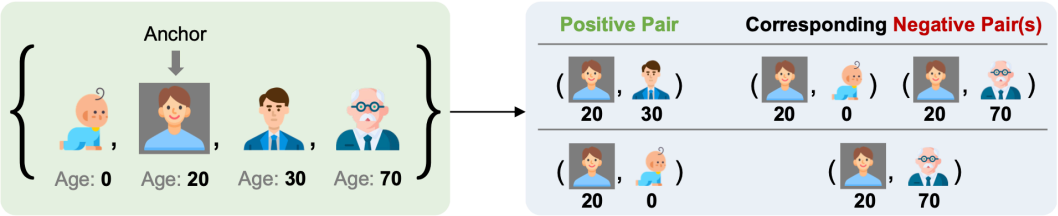


Illustration of the RNC framework. Figure from Zha et al. [2]

Encoder-Predictor Framework: $x \xrightarrow{\text{encoder } f(\cdot)} v \xrightarrow{\text{decoder } g(\cdot)} \hat{y}$

Core Idea: Enforce an ordering constraint in the feature space that matches the label space ordering. Given input-label pairs $\{(x_n, y_n)\}_{n\in[N]}$, we aim to learn the latent space embeddings $v_\ell = f(x_\ell) \in \mathbb{R}^{d_e}$ respecting the following constraint:

Order Constraint:

$$d(y_i, y_j) < d(y_i, y_k) \Rightarrow \mathsf{sim}(v_i, v_j) > \mathsf{sim}(v_i, v_k)$$

Rank-N-Contrast (RNC): Contrastive Learning Framework

1. Rank-Based Sample Grouping

- For an anchor embedding v_i from sample $(\tilde{x}_i, \tilde{y}_i)$, define a rank-based negative set:

$$S_{i,j} := \{v_k \,|\, k \neq i,\ d(\tilde{y}_i, \tilde{y}_k) \geq d(\tilde{y}_i, \tilde{y}_j)\}$$

2. Exponential Likelihood and RNC Loss

- The likelihood that v_j is more similar to v_i than the elements in $S_{i,j}$ is given by:

$$P(v_j \mid v_i, S_{i,j}) = \frac{\exp\left(\mathrm{sim}(v_i, v_j)/\tau\right)}{\sum_{v_k \in S_{i,j}} \exp\left(\mathrm{sim}(v_i, v_k)/\tau\right)}$$

- For each anchor v_j, we want to maximize $P(v_j \mid v_i, S_{i,j}) \Leftrightarrow$ minimize the negative per-sample RNC loss (sum over all elements as anchors in the batch):

$$\ell_{\mathrm{RNC}}^{(i)} = \frac{1}{N-1} \sum_{\substack{j=1 \\ j \neq i}}^{N} -\log P(v_j \mid v_i, S_{i,j})$$

RNC-inspired Contrastive Learning for POSEIDON : Key Challenges

- **Architectural Complexity:** 21M parameter scOT transformer
- **Feature Dimensionality:** 12k-dimensional embeddings
- **Target Space Complexity:** 2D vector fields (u, v, ρ) vs. scalar targets in standard contrastive regression

Proposed Solutions and Contributions:

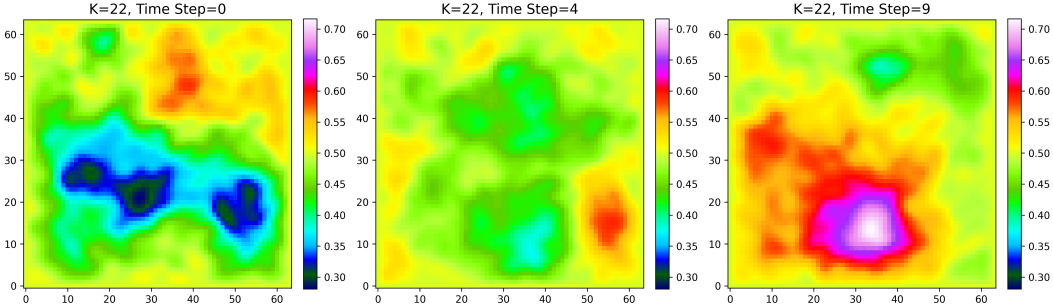
1. **Two-Stage Dimensionality Reduction:** $\mathbb{R}^{12288} \xrightarrow{\text{GELU}} \mathbb{R}^{1024} \xrightarrow{L_2 \text{ norm}} \mathbb{R}^{512}$
2. **Additional Reconstructive Loss:** $\mathcal{L}_{\text{total}} = (1-\alpha)\mathcal{L}_{\text{RNC}} + \alpha\mathcal{L}_{\text{recon}}$
3. **Temperature Annealing:** Temperature parameter τ decreasing (cosine decay) with training steps
4. **Dataset Creation and Performance Metrics:** Datasets (Wave Equation; Navier-Stokes Equation) compatible with all2all training; performance metrics using both the baseline regression loss L_1 and the contrastive approach

Wave Equation Dataset: $\partial_{tt} u - c^2 \Delta u = 0$, $c = 0.1$

Initial Condition: $u_0(x, y) = \frac{\pi}{K^2} \sum_{i,j=1}^{K} a_{ij} (i^2 + j^2)^{-r} \sin(\pi i x) \sin(\pi j y)$

Analytical Solution:

$$u(x, y, t) = \frac{\pi}{K^2} \sum_{i,j=1}^{K} a_{ij} (i^2 + j^2)^{-r} \sin(\pi i x) \sin(\pi j y) \cos\left(c \pi t \sqrt{i^2 + j^2}\right)$$

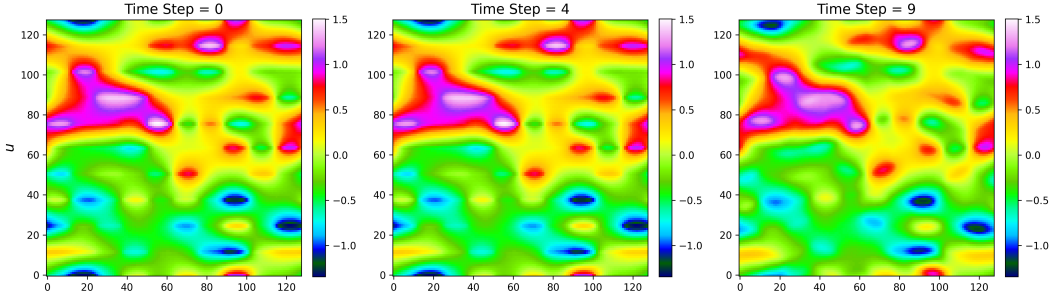


Wave equation. Analytical solution u for $K = 22$ at different time steps.

Navier-Stokes Equation: $\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} + \nabla p = \nu \Delta \mathbf{u}$, $\nabla \cdot \mathbf{u} = 0$

Initial Conditions:

- **NS-Sines:** Velocity fields defined via sine/cosine modes
- **NS-Gauss:** Superposition of smooth Gaussians in vorticity
- **NS-PwC:** Piecewise constant vorticity



NS-PwC. Numerical approximation u of the incompressible Navier-Stokes equation at different time steps between $t = 0$ and $t = 5$. Initial condition of type NS-PwC.

Experimental Setup and Training Configuration

Wave Equation Dataset:

- **ID:** $K \in [18, 25]$ Fourier modes
- **OOD:** $K \in [26, 28]$ Fourier modes
- 1024 training trajectories
- 10 time steps $\rightarrow$ 45,000 training pairs

Navier-Stokes Dataset:

- **Training:** NS-Sines + NS-Gauss
- **Zero-shot:** NS-PwC
- 2×1024 training trajectories
- 7 time steps $\rightarrow$ 25,000 training pairs

Training Approach:

- **One-stage training:** Combined loss eliminates two-stage approach
- **Larger batch sizes:** Up to 512 vs 32 for L_1 regression
- **Temperature parameter:** Temperature Annealing or Fixed $\tau = 1.75$ (critical for convergence)
- **Similarity measure:** Negative L_2 norm outperforms cosine similarity
- **Reconstruction loss:** Provides training stability in early iterations; serves as a feedback for the contrastive loss

Main Results: Evaluation Metrics

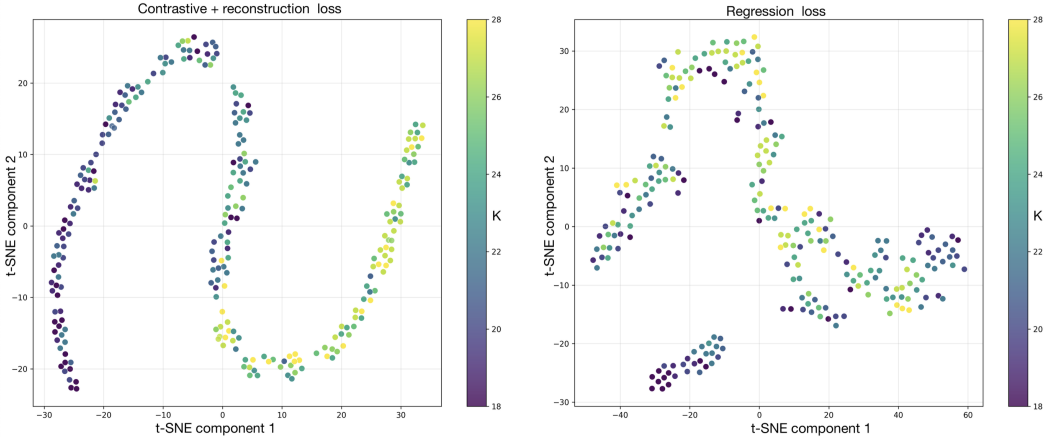
Wave Equation Dataset Performance:

	In-distribution			Out-of-distribution		
Method	**Avg**	**Min**	**Max**	**Avg**	**Min**	**Max**
POSEIDON-T L_1	2.03	0.77	6.26	1.44	0.80	4.08
POSEIDON-T $\mathcal{L}_{\text{RNC}}$	1.34	0.50	5.99	0.91	0.49	3.91
Improvement	**34%**	**35%**	**4%**	**37%**	**39%**	**4%**

Navier-Stokes Equations Performance:

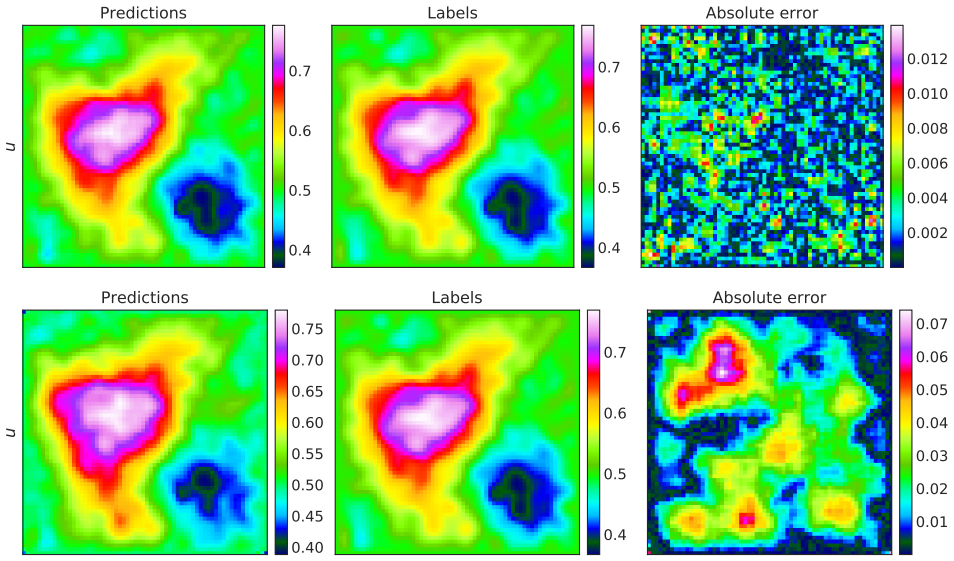
	NS-Gauss	NS-Sines	NS-PwC (zero-shot)		
Method	**(ID)**	**(ID)**	**Avg**	**Min**	**Max**
POSEIDON-T L_1	2.62	3.43	2.34	1.61	3.31
POSEIDON-T $\mathcal{L}_{\text{RNC}}$	2.19	3.25	2.13	1.21	3.28
Improvement	**16%**	**5%**	**9%**	**25%**	**1%**

Main Results: Latent Space Structure

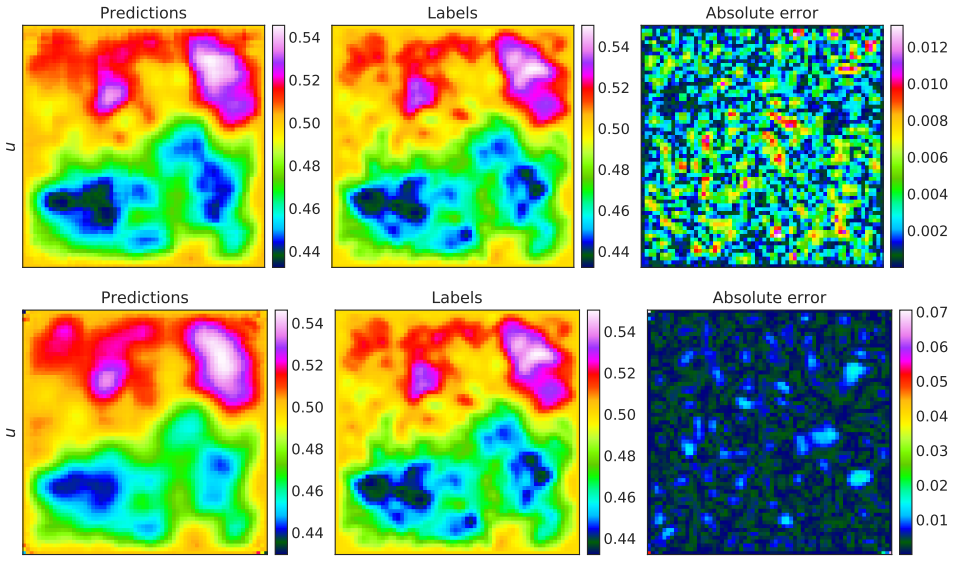


Learned representations on the Wave equation dataset. The regression learning (L_1) (right) fails to recognize the underlying continuous information in the data. The contrastive approach (left) captures the intrinsic continuity in the K parameter.

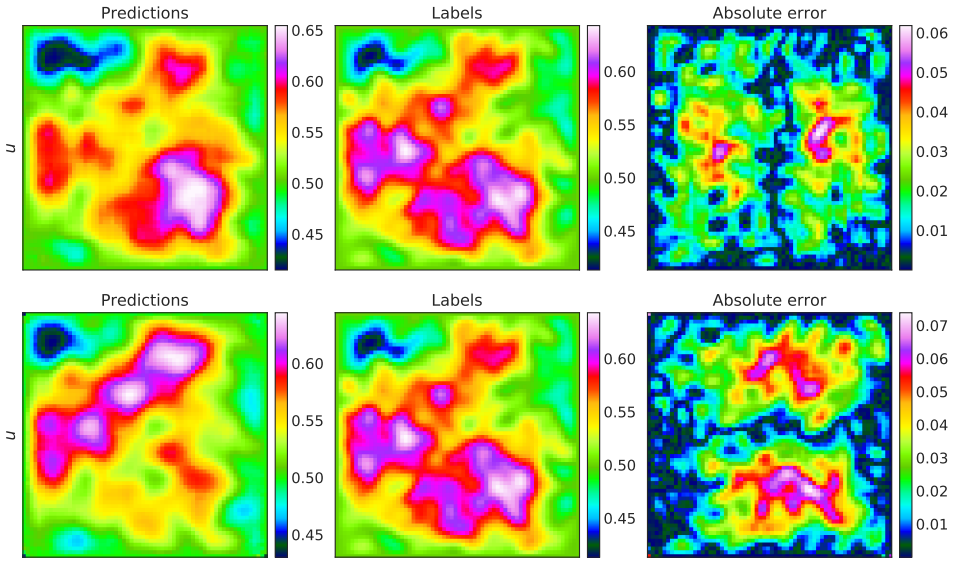
Model Predictions: Wave Equation Comparison (cont. on top) - ID



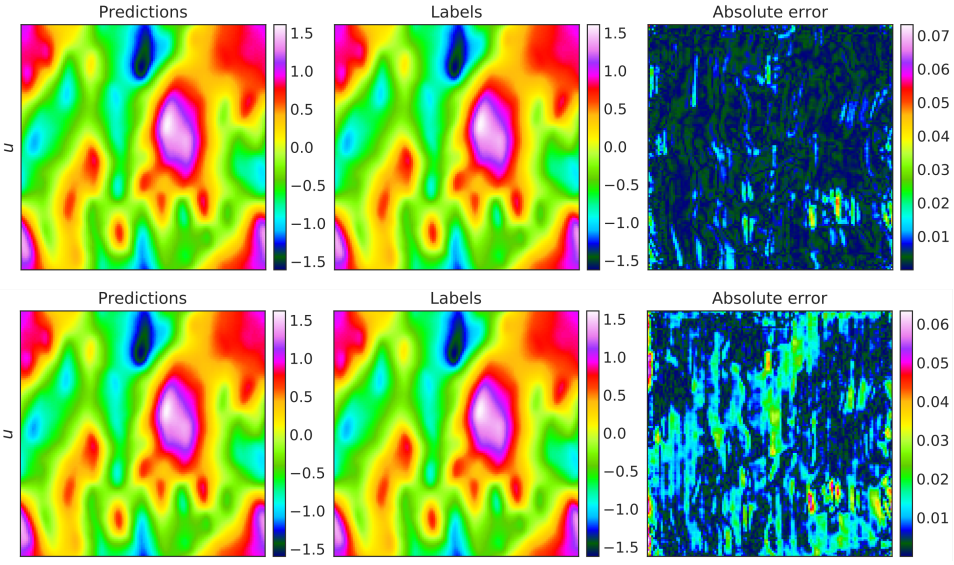
Model Predictions: Wave Equation Comparison (cont. on top) - OOD



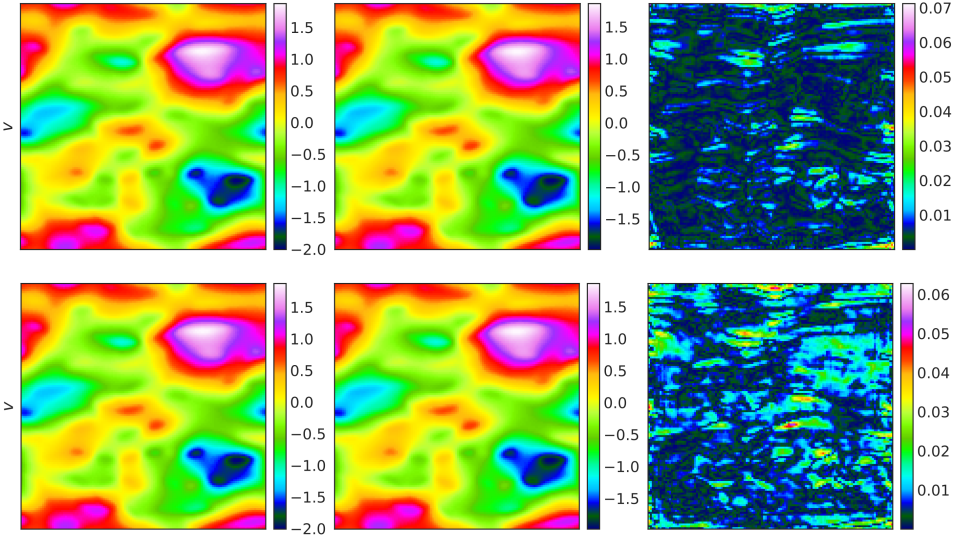
Model Predictions: Wave Equation - Worst Predictions (cont. on top)



Model Predictions: NS Equation (cont. on top) - PWC zero-shot - u



Model Predictions: NS Equation (cont. on top) - PWC zero-shot - v



Conclusion

Implemented a contrastive learning framework within the POSEIDON architecture.

Performance Results

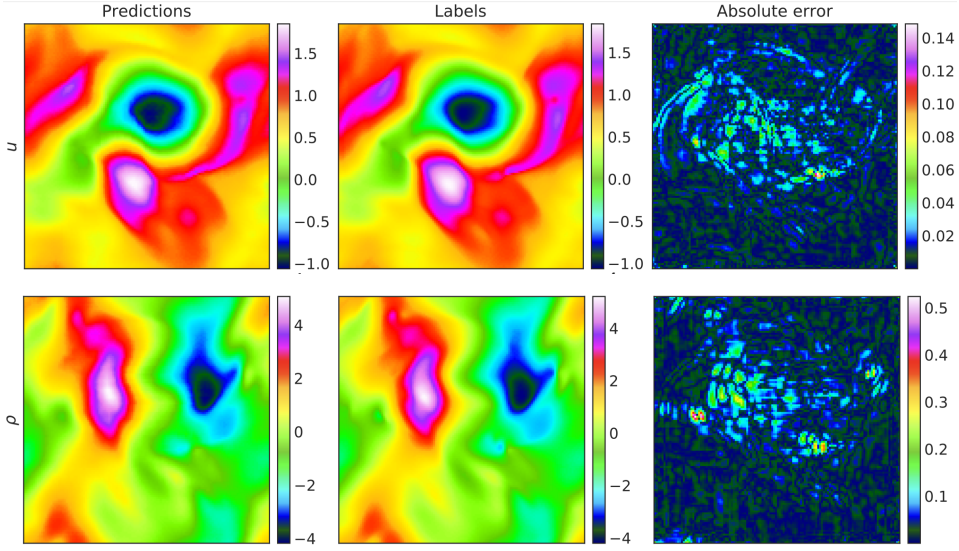
Achieved superior performance in both in-distribution and out-of-distribution settings on time-dependent fluid dynamics datasets:

- Achieved increased accuracy and reduced training time compared to standard regression loss.
- Latent space shows improved continuity and better reflects underlying physical parameters.

Key Insights

- Demonstrated the ability of contrastive learning to uncover complex flow structures and physical relations that are overlooked by standard approaches.
- This suggests contrastive learning as a promising direction for improving foundation models like POSEIDON in learning PDE solution operators.

Outlook: More complex datasets + Benchmarking



References

[1] Herde, M., Raonic, B., Rohner, T., et al. (2024). POSEIDON: Efficient Foundation Models for PDEs. *arXiv preprint arXiv:2405.19101*.

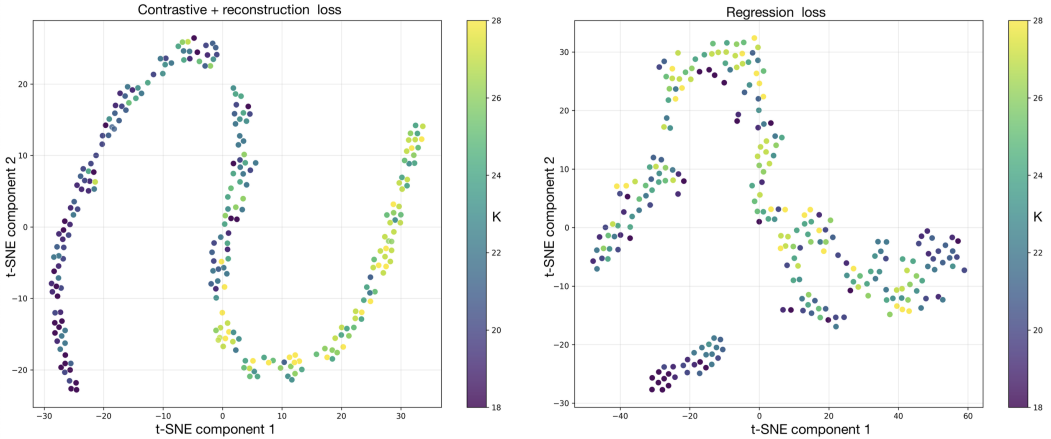
[2] Zha, K., Cao, P., Son, J., Yang, Y., & Katabi, D. (2022). Rank-N-Contrast: Learning Continuous Representations for Regression. *Advances in Neural Information Processing Systems*.

[3] Mishra, S. (2024). Numerical Analysis meets Machine Learning. *Foundations of Computational Mathematics*.

[4] Bommasani, R., Hudson, D. A., Adeli, E., et al. (2022). On the Opportunities and Risks of Foundation Models. *arXiv preprint arXiv:2108.07258*.

[5] Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A Simple Framework for Contrastive Learning of Visual Representations. *Proceedings of the International Conference on Machine Learning*.

Thank you for your attention!



Learned representations on the Wave equation dataset. The regression learning (L_1) (right) fails to recognize the underlying continuous information in the data. The contrastive approach (left) captures the intrinsic continuity in the K parameter.

Additional Material: all2all Training Strategy

Theorem (Temporal Consistency Property)

For any PDE solution $u(t) = \mathcal{S}(t, a)$*, we have:*

$$u(t') = \mathcal{S}(t' - t, u(t)) \text{ for any } 0 \leq t < t' \leq T$$

all2all Loss Function:

$$\widehat{\mathcal{L}}(\theta) = \frac{1}{M\widehat{K}} \sum_{i=1}^{M} \sum_{\substack{k,\bar{k}=0 \\ k<\bar{k}}}^{K} \left\| \mathcal{S}\left(t_{\bar{k}} - t_k, u_i(t_k)\right) - \mathcal{S}_\theta^*\left(t_{\bar{k}} - t_k, u_i(t_k)\right) \right\|_{L^p(D)}^p$$

where $\widehat{K} = \frac{(K+1)(K+2)}{2}$ and $u_i(t_k) = \mathcal{S}(t_k, a_i)$.